

Online Coverage Planning for an Autonomous Weed Mowing Robot With Curvature Constraints

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Abstract—The area used for grazing cattle constitutes about one-third of the land in United States. In this paper, we present the design of Cowbot, an autonomous weed mowing robot for maintaining cow pastures. Cowbot is designed to operate in rugged environments and provides a cost-effective method for weed control in cow pastures. Path planning for the Cowbot is challenging since weed distribution on pastures is unknown. Given an onboard sensor with a limited field of view, online planning is necessary to detect weeds and plan paths to mow them. We study the general online coverage planning problem for an autonomous mower with curvature and field of view constraints. We develop two algorithms that are able to utilize new information on weed detection to optimize path length and ensure coverage. We deploy our algorithms on the Cowbot and perform field experiments to validate the suitability of our methods for real-time path planning. We also perform extensive simulation experiments which show that our algorithms result in up to 60% reduction in path length as compared to boustrophedon and random-search based coverage paths.

Index Terms—Agricultural automation, field robots, motion and path planning, nonholonomic motion planning.

I. INTRODUCTION

DAIRY pastures constitute about one third of the land in the United States [1]. A single dairy farm averages around 460 acres. Maintaining these large pastures to prevent weeds from taking over the nutritious grasslands is a daunting task, especially in organic farms that do not use herbicides. Many recent projects have developed robotic weeding solutions for outdoor [2]–[5] and indoor (greenhouse) [6], [7] weed control applications in agriculture. However, robotic weed removal efforts to-date have primarily focused on crop-cultivating farms that practice row-plantation. By virtue of row plantations, the robots have a strong prior on the outlay of the field and the location of weeds - inter-row and intra-row. Cow pastures on the other hand are usually large fields with little to no regular maintenance. In the case of organic dairy farms, that form the focus group for this work, weed growth is significant and can spread fast if not controlled in time. Rough uneven terrain combined



Fig. 1. Cowbot - autonomous weed mower operating on a cow pasture. As shown in the image on the right, weeds on cow pastures can grow up to 1-2 meters tall.

with large and dense weed populations makes it challenging to operate in cow pastures. This results in the need to develop a robust and intelligent robotic weeding platform.

In this paper, we present the design of our autonomous weed mowing robot for cow pastures called Cowbot (Fig. 1). Cowbot is designed to operate in the rugged environments on cow pastures and provides a cost-effective method for weed control in organic fields. The density of weeds on a pasture affects the cost of operation of the Cowbot. In fields with dense weed cover, a boustrophedon coverage path [8] is usually the most efficient path to mow all weeds. In case of low density or localized distribution of weeds, a boustrophedon path results in considerable wastage of resources, specially on large fields. We assume a finite field of view of the Cowbot within which it is able to detect and localize all weeds. Then, an intuitive solution is to perform a random search on the pasture and mow weeds as they are detected. This strategy would eventually allow the Cowbot to mow all weeds but as shown in Section VII, it is not cost effective since the random search results in repetitive coverage of same areas. Another major drawback to any random search based method is not knowing when to terminate. Common termination criteria include stopping when no new weeds are detected for a period of time or when a threshold for maximum path length (or time) is reached. This necessitates the design of intelligent path planning algorithms to ensure all weeds on the pasture are detected and mowed by the autonomous mower. We refer to the online path planning for Cowbot as the mower routing problem (MRP). In this paper, we present a formulation for MRP and develop two online path planning algorithms, called JUMP and SNAKE, to compute efficient solutions. Our path planning algorithms use new information about weeds on the pasture to improve path length by up to 60 % as compared to boustrophedon and random-search based coverage paths. We implement our algorithms on the Cowbot and perform field experiments to validate that our

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algorithms are suitable for real-time path planning. We also perform large-scale computational simulations and benchmark the performance of our solution methods with online and offline baseline methods. We show that JUMP provides 100% weed coverage while trading off in terms of path length as the number of weeds increases. SNAKE is best suited for low density weed populations and R-SNAKE (a variant of SNAKE) provides an efficient alternative when 100% weed removal is not essential.

II. RELATED WORK

The design and development of mobile robots for weed removal applications has witnessed growth in recent years. Bakker *et al.* [9] have designed a weeding robot for intra-row weed removal. Bawden *et al.* [2] develop a versatile agricultural robot equipped with a heterogeneous set of weeding implements and develop weed detection methods. Grimstad and From [10] present the design of Thorvald agricultural robot and demonstrate the robustness and modularity of their design in field trials. While there exists literature on weed detection and system design, there is relatively little work on path planning for agricultural robots beyond coverage path planning. Weed distribution on pastures is not known a priori and is hard to estimate. As a result boustrophedon-like coverage paths can lead to large wastage of resources.

The mower routing problem finds relevance in many well-studied problems such as variants of the traveling salesman problem (TSP), coverage path planning (CPP) and curvature constrained path planning. In dynamic TSP [11] for instance, the distribution of city locations is unknown at the start of the problem. However, unlike in MRP wherein the mower must search for weeds on the pasture, city locations in dynamic TSP are generated by a stochastic process and made available to the salesman during path execution. Another variant of TSP that requires the path to satisfy non-holonomic constraints of a Dubins motion model [12] is called Dubins TSP [13]. While both of these problems are similar to MRP, they do not capture all of its constraints. Hence their solution methods cannot be used to solve MRP. Coverage path planning also relates to MRP and addresses coverage of bounded spaces with a finite sensor footprint. Galceran and Carreras [14] and more recently Khan *et al.* [15], present literature reviews on CPP variants and their solutions. Khan *et al.* [15] focus on CPP methods for curvature constrained mobile robots. Online variants of CPP such as active [16]–[19] and sensor-driven [20]–[23] coverage path planning also find relevance to our work. The primary focus of this line of research is to ensure that utility of data collected by the robot(s) is maximized based on application specific metrics. These application areas include classification [18], [19], 3D reconstruction [16], [17], aerial coverage [22], resolution-complete coverage [23], [24], inspection to detect mines and/or chemicals [21], seabed monitoring [17], [20] and ship hull inspection [16].

However, none of these problems and their solution methods extend to MRP. In the case of MRP, in addition to online and curvature-constrained coverage of the environment with its sensing footprint, the robot must also visit each weed location

detected in the field using a different coverage footprint (mowing footprint). Existing work in the literature does not address this coupled coverage requirement. A solution to MRP would be useful in other similar applications such as detecting and collecting trash [25], identifying and collecting high utility rock samples for planetary rovers [26] and detecting & removing rocks from agricultural fields. Existing work on these problems only considers a subset of constraints of MRP and does not deploy their algorithms on actual robots. For instance, [25] considers a fully actuated robot model that does not have any curvature constraints for trash picking and [26] addresses an offline problem that assumes that the location of all sample sites is available to the planetary rover a priori. There does not exist any work in the literature that addresses MRP.

In this paper, we present the design of our autonomous weed mowing robot for cow pastures called Cowbot. We formalize the online coverage planning problem for the Cowbot to mow weeds as the mower routing problem and develop two online solution methods, called JUMP and SNAKE. We compare the performance of our online planning methods with coverage and TSP based solution methods. We also compare our methods with a random-search based reactive online planner. We conduct extensive simulations and show that our algorithms lead to up to 60% shorter paths as compared to boustrophedon or random-search based methods by efficiently utilizing information about weed locations. We also deploy our algorithms on the Cowbot and conduct field experiments to verify computational efficiency and suitability of our methods for real-time path planning.

Paper Organisation: We present the system design for Cowbot in Section III and formalize the problem statement for MRP in Section IV. In Section V, we discuss preliminaries and develop upper and lower bounds for the solution space. We present our solution methods for MRP in Section VI. In Section VII and VIII, we present results from simulations and field experiments, respectively. In Section IX we discuss conclusions and identify future directions to this work.

III. COWBOT SYSTEM DESIGN

Cowbot is built on the Toro Groundmaster 3280-D rotary mower platform. It has a four-wheel drive transmission system with steering control on the rear wheels and a turning radius of 1.5 m at the geometric center. It weighs around 800 kgs. It is customized to operate with a 32 kW electric motor powered by a 28.8 kWh lithium ion battery pack. It is designed to be a standalone solution for weed control. As part of the project, a cargo trailer was equipped with retractable solar panels on the roof and climbing ramps to house the Cowbot and recharge it on the pasture without the need for any charging infrastructure. Cowbot incorporates a suite of safety features including a wireless remote emergency stop button, pressure activated perimeter bumpers and emergency stop knobs on the front and back of the mower. It comprises a flail-deck weeding implement to reduce the risk of flying debris during autonomous operations. The flail-deck has a cutting width of 1.3 m and weighs 225 kgs. It is custom mounted with high ground clearance to achieve a cut-height of

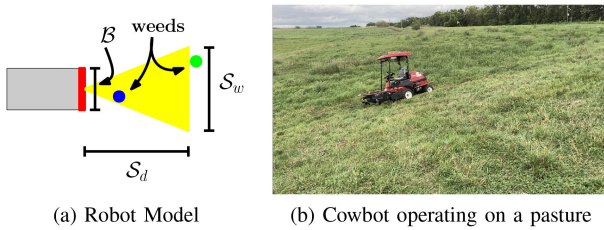


Fig. 2. (a) Robot model showing the robot (gray), detection field of view (yellow), weeding implement (red) and weeds (solid disks) in the pasture. The green disk shows an undetected weed on the pasture and blue disk shows a weed detected by the mower within the detection field of view. (b) Cowbot operating on steep slopes in the pasture.

20 cm (8 in.). The high cut ensures sufficient volume of grass for the cows even after mowing while also stunting weed growth.

Cowbot uses a GPS-based point to point inertial navigation system and is equipped with two multi-band RTK GNSS receiver units (SwiftNav Duro Inertial) with inbuilt IMU sensors to provide high accuracy location and heading estimates. It is also equipped with front facing imaging sensors that includes two RGB-D cameras (Intel RealSense D455) and a lidar sensor (Velodyne Puck VLP-16). The perception stack of the Cowbot that aims to build capability for visual inertial navigation and weed detection on the field is currently under development. The sensors connect to an onboard laptop computer (Dell Precision 7530) that runs the navigation software. The navigation software on Cowbot has a three-layer architecture. The top and middle layers execute on the laptop and the bottom layer executes on a programmable logic controller (TORO PLC model # - Tec 5004) that controls the traction and steering systems. The top layer is the perception and planning layer that takes input from the user and on-board sensors, and plans paths for the mower. The middle layer comprises of high level control modules for point to point navigation. It communicates with the top layer over the robot operating system (ROS) and takes as input the waypoint path computed by the planner. It generates steering and speed commands for the bottom layer. The bottom layer generates signals for the electric motor and the hydraulic steering to follow commanded speed and steering values. The middle and bottom layers communicate using CANBUS messaging protocol.

IV. PROBLEM FORMULATION

In this section, we define the system model for the Cowbot used in path planning and present the problem statement.

A. System Model

As the Cowbot moves in the pasture, it sweeps the ground with the weeding implement and mows any weeds that it passes over. For the purpose of path planning, we approximate its motion model with non-holonomic motion constraints that follow Dubins car-like vehicle model [12] and moves with a constant forward speed. It has a finite turning radius, $\mathcal{R}$, and similar to the Cowbot is equipped with a front mounted weeding implement (see Fig. 2) that it pushes as it moves forward. The weeding

implement has a finite width, $\mathcal{B}$. A weed is considered mowed when the implement passes over it.

We also consider the mower model to consist of a weed detection module that includes a suite of front facing imaging sensors and continues to operate throughout the execution of a path. The weed detection field of view (FOV) is finite, triangular shaped and apexed at the front center of the mower (see Fig. 2(a) for an illustration of the FOV). The depth, S_d , and width at the farthest point in the FOV, S_w , define the range of the detection module on the mower. The mower is able to detect and localize a weed in the global frame of reference instantaneously as the weed enters its FOV.

B. Problem Statement

Let $\mathcal{E}$ be a rectangular pasture. $\mathcal{E}$ is assumed to be obstacle-free and fully traversable by the mower. The mower does not have any prior information on the location or distribution of weeds within $\mathcal{E}$. To ensure all weeds on the pasture are detected, the mower must cover the entire pasture with its FOV and perform path planning to visit and mow each of the detected weeds. The *Mower Routing Problem (MRP)* can then be defined as follows:

Given a pasture $\mathcal{E}$ with unknown weed distribution and an autonomous mower equipped with a front mounted weeding implement and a finite FOV (as defined in Section IV-A), develop an online path planning policy that plans a path for the mower to detect and mow all weeds in $\mathcal{E}$ while minimizing the total length of the path.

V. PRELIMINARIES

In this section, we define the notation used to develop the solution methods, discuss Dubins path preliminaries and compute bounds on the solution space to MRP.

Given a rectangular pasture $\mathcal{E}$ of size $(L \times W)$, we consider an inertial frame of reference such that the four corners of the pasture are located at $(0, 0)$, $(L, 0)$, (L, W) and $(0, W)$. For ease of exposition we will refer to locations with bigger y -coordinates as being located above locations with smaller y -coordinates, and vice-versa as being located below. Current state of the mower is represented as (x_m, y_m, θ_m) , where (x_m, y_m) are the location coordinates and θ_m is the heading angle. The mower maintains a list, $\mathcal{W}$, of weeds detected on the pasture that are yet to be mowed. Each weed is marked as a point object on the field and is identified by its location coordinates, $w_i = (x_i, y_i)$.

A. Dubins Paths

We use primitives from Dubins shortest path algorithm for curvature constrained vehicles [12] to compute traversable paths for the Cowbot. Specifically we consider Dubins paths of the type C^+SC^- , where C^+ and C^- are circular arcs in opposite direction and S is a straight line path. LSR and RSL paths are realizations of the C^+SC^- path type, where R refers to a circular arc in the clockwise direction and L refers to a circular arc in the counter clockwise direction. Both circular arcs turn with the minimum turn radius, $\mathcal{R}$.

B. Bounding the Solution Space

Weed distribution on the pasture significantly affects the length of the mower's path needed to mow all weeds. We present two edge-case scenarios and use them to compute upper and lower bounds on the solution space.

First, consider a pasture with a uniform weed distribution and densely packed weeds. The mower must mow every point on the pasture to ensure that all weeds are mowed. An optimal mower path in this scenario is a boustrophedon coverage path (BCP). A boustrophedon path [8] to cover the pasture comprises of straight line passes of length L across the pasture that are parallel to x -axis. Consecutive passes in the boustrophedon path have opposite direction of motion and are B distance away from each other. Let $y_p(i)$ be the y -coordinate corresponding to the i^{th} pass in the path and $\theta_p(i)$ be the direction of motion on the pass. The first pass of the path is along the line, $y_p = B/2$ and the last pass is along the line $y_p = \lceil \frac{W}{B} \rceil \times B - B/2$. When it is clear from the context we use y_p and θ_p interchangeably with $y_p(i)$ and $\theta_p(i)$, respectively. Such a path ensures all weeds on the pasture are mowed regardless of the weed distribution and provides an upper bound on the path length for the mower.

Second, consider the scenario with no weeds on the pasture. Since the mower does not have any information about the weed distribution, it must at least cover the pasture with its FOV to search for weeds. Consider another BCP but with spacing between consecutive passes equal to S_w . This reduces the total number of passes by a factor of $\frac{S_w}{B}$. At the end of this path, the mower has deterministic knowledge of all weed locations (or of the absence of weeds) on the pasture. This is the minimum length path needed for the mower to search for weeds on the pasture and in the case no weeds are detected it forms the solution to the MRP instance. This provides a lower bound on the solution space of MRP.

VI. SOLUTION METHODS

In this section, we present solution methods to the mower routing problem to detect and mow weeds on a pasture. We design two online planning algorithms, named JUMP and SNAKE, that build on top of BCP. We present systematic modifications to BCP that account for Cowbot's curvature constraints and utilize its FOV and width of the weeding implement to enable efficient online coverage of the pasture.

A. JUMP Online Planning Algorithm

The JUMP planner modifies BCP by introducing the ideas of *jumps* and *spring*. It computes a path for the mower that starts at the bottom of the pasture and makes its way towards the top while doing to-and-fro passes across the pasture much like a boustrophedon path. A *jump* allows the mower to make detours to mow nearby weeds within its FOV and then return back to the current pass to continue on its path. *Spring* behavior of the JUMP planner permits variable spacing between consecutive passes, unlike BCP, resulting in a stretch and shrink pattern to better utilize new information about weeds on the pasture.

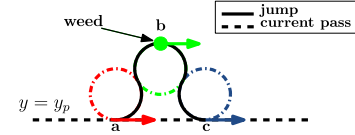


Fig. 3. LSR-RSL jump to mow a weed at point (b) with $\theta_p = 0$. (a) and (c) are the start and end points of the jump.

When there are no weeds directly in front of the mower on its current path, it makes a search for nearby weeds that it can mow by making a quick detour. We restrict these detours to weeds that are located above the current pass, thus leading to the name *jump*. A jump consists of two *sub-paths*: first, to go *up* to the weed's location from the current position and second, to come back *down* to the current pass. We use Dubins paths of type RSL and LSR to compute sub-paths for a jump. Each jump is defined by three waypoints: start and end points of the jump and the location of the weed. We fix the heading angle at each of the three waypoints to θ_p . When $\theta_p = 0$, jumps comprise of sub-path LSR to go to the weed and RSL to come back to the path (Fig. 3), and vice-versa when $\theta_p = \pi$. Ties among jumps starting at the same point are broken arbitrarily.

At the end of a pass, the planner must compute $y_p(i+1)$ value for the next pass with $\theta_p(i+1) = \pi - \theta_p(i)$. It is computed as the minimum of the three expressions in (1).

$$y_p(i+1) = \min \begin{cases} y_p(i) + S_w/2 \\ \min_{\forall i: w_i \in \mathcal{W}} y_i + B/2 \\ W - B/2 \end{cases} \quad (1)$$

JUMP planner admits dynamic spacing between consecutive passes allowing a *spring*-like behavior. To this end, the first condition corresponds to maximum spacing (equal to $S_w/2$), the second condition corresponds to the bottom-most unmowed weed and admits tighter spacing (*shrink*), and the third condition corresponds to the pass along the top edge of the pasture. Note that, the first condition permits for $y_p(i+1) < y_p(i)$ thus allowing the mower to go in the negative y -direction on the next pass. This occurs when the mower detects a new weed that lies below its current pass.

JUMP planner maintains the following two invariants throughout its execution.

Invariant 1: At the start of a pass there does not exist any unmowed weeds in $\mathcal{W}$ below $(y_p - B/2)$.

Proof: The invariant is true on the first pass by virtue of it being at the bottom edge of the pasture. In consecutive passes, the computation of y_p ensures coverage of the bottom most weed in $\mathcal{W}$, thus maintaining the invariant. ■

Invariant 2: Consider $\mathcal{W}_{start}$ to be a snapshot of $\mathcal{W}$ at the start of a pass. At the end of the pass, there does not exist any unmowed weeds in $\mathcal{W}_{start}$ below $(y_p + B/2)$.

Proof: If the mower traverses a pass without any jumps, all weeds in $\mathcal{W}$ below $y_p + B/2$ are mowed. The planner computes a jump for the mower only in the case when there are no weeds in $\mathcal{W}$ on the current pass that will be skipped as a result of the jump. This maintains the invariant at the end of each pass. ■

The two invariants along with the termination criteria (2), ensure that the mower mows all weeds detected on the pasture.

$$(y_p == W - \mathcal{B}/2) \cap (\mathcal{W} == \phi) \quad (2)$$

Termination criteria for JUMP is evaluated at the end of each pass and evaluates to true if the mower has reached the top of the pasture and there are no unmowed weeds left in $\mathcal{W}$.

B. SNAKE Online Planning Algorithm

The SNAKE planner derives its name from serpentine to-and-fro patterns across the pasture generated in paths computed by the planner. It builds on top of the JUMP planner by allowing detours in both directions. However, unlike JUMP, SNAKE does not require the mower to return to its current pass after mowing a weed, thus only completing the first sub-path on a jump. A pass of the SNAKE path across the pasture often comprises of a sequence of sub-paths to weed locations. SNAKE enforces a consistent direction of motion on a pass by fixing the heading angle at each weed location to θ_p .

At the start of a pass, the planner makes a search for the closest weed in $\mathcal{W}$ that lies ahead of the mower along its direction of motion. If $\theta_p = 0$, it considers Dubins paths of type LSR and RSL to compute sub-paths to weeds located above and below y_m respectively, and vice-versa when $\theta_p = \pi$. While the mower is executing a sub-path, the planner does not search for new sub-paths. It resumes the search when the mower completes a sub-path. If there do not exist any weeds in $\mathcal{W}$ along the mower's direction of motion, the mower continues to move straight ahead with heading angle $\theta_m = \theta_p$. When the mower reaches the end of the pasture along its direction of motion, the planner computes $y_p(i+1)$ value (using (3)) for the start of the next pass with $\theta_p(i+1) = \pi - \theta_p(i)$.

$$y_p(i+1) = \min \left\{ \begin{array}{l} y_p(i) + \mathcal{S}_w/2 + \mathcal{B}/2 \\ W - \mathcal{B}/2 \end{array} \right. \quad (3)$$

The value for $y_p(i+1)$ is computed as the minimum of the two expressions in (3). The first expression corresponds to fixed spacing between passes (equal to $\mathcal{S}_w/2 + \mathcal{B}/2$) and the second expression corresponds to the pass starting along the top edge of the pasture. Note that, in the case of SNAKE, y_p only determines the start point of a pass. The mower typically veers off of y_p while executing sub-paths to mow weeds. It continues traversing on consecutive passes across the pasture alternating in its direction of motion until the termination criteria is met. Termination criteria for SNAKE ((4)) is evaluated at the end of each pass. It evaluates to true if the mower has reached the top of the pasture and there are no unmowed weeds left in $\mathcal{W}$. The mower is determined to have reached the top of the pasture if the current pass starts along the top edge of the pasture i.e. $y_p = W - \mathcal{B}/2$, and the mower's current location is also along the top edge i.e. $y_m \geq W - \mathcal{B}$.

$$(y_m \geq W - \mathcal{B}) \cap (y_p == W - \mathcal{B}/2) \cap (\mathcal{W} == \phi) \quad (4)$$

We also define a variation of the SNAKE planner called RESTRICTED-SNAKE (R-SNAKE) that trades off weed coverage in favor of shorter path length for the mower. While SNAKE

admits sub-paths to all unmowed weeds along its direction of motion, R-SNAKE restricts its search to weeds, $w_i \equiv (x_i, y_i)$, located at most a fixed threshold below the y_p value of the current pass i.e.:

$$y_i \geq y_p - \frac{3}{2}\mathcal{S}_w \quad (5)$$

R-SNAKE uses the termination criteria given in (6) and evaluates it at the end of each pass. As is apparent from the termination criteria, R-SNAKE can potentially miss some weeds on the pasture.

$$(y_m \geq W - \mathcal{B}) \cap (y_p == W - \mathcal{B}/2) \quad (6)$$

VII. SIMULATION RESULTS

To evaluate the performance of the online planning algorithms presented in Section VI we designed a simulated pasture environment and an autonomous mower operating on the pasture. The simulator was implemented in Python 3 programming language and uses PyDubins [27] and Elkal [28] libraries to compute Dubins paths and TSP tours, respectively. The simulations were run on a desktop computer with AMD Ryzen 7 processor and 32 GB RAM. Psuedocode for our algorithms is available at [29]. We compare the performance of our online algorithms with three baseline solutions. First, the boustrophedon coverage path (BCP) used to compute upper bounds on the path length for MRP as given in Section V-B. We recall that path length in case of BCP is independent of the weed density and spatial distribution and is determined by the size of the pasture and mower's parameters. The second baseline called BCP-TSP comprises of two phases. The first phase involves visual coverage of the pasture using the lower bounding BCP (as described in Section V-B), while also mowing any weeds that lie directly on the mower's path. The second phase computes a TSP tour for the mower to visit unmowed weeds detected in the first phase. The third baseline is an online reactive planner (REACT) that performs a random search for weeds on the pasture and mows weeds in the order they are detected. The planner generates random waypoints for the mower to perform a search. When it detects a weed, the mower aborts the random search and starts to visit weed locations in FIFO order while still detecting new weeds that enter the mower FOV. It resumes the random search once all detected weeds are mowed. It terminates when the path length equals BCP path length (upper bound).

A. Simulation Setup

We perform the simulations in a pasture of size 100 m $\times$ 40 m. To populate weeds on the pasture we use two different spatial distributions, uniform (U) and Gaussian (G). In the case of Gaussian distribution, 20% of the weeds are sampled uniformly at random and chosen as seeds for Gaussian clusters used to generate rest of the weed locations. Fig. 4 shows simulation instances with uniform (Fig. 4(b), (c) and (d)) and Gaussian (Fig. 4(f), (g) and (h)) weed distributions and paths planned by the different planners. The mower is configured to operate with a constant speed of 1 m/s and comprises of a front mounted weeding implement of width $\mathcal{B} = 2m$. We vary the number of

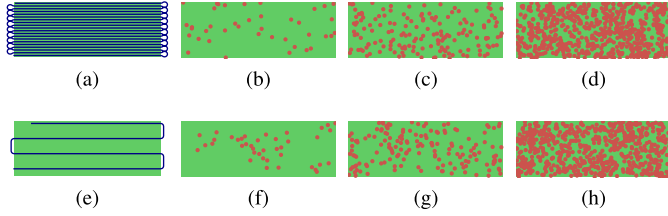


Fig. 4. (a) upper bounding BCP path. (e) lower bounding BCP path. (b)-(d) show uniform distribution of weeds on a simulated pasture with 40, 160 and 640 weeds respectively. (f)-(h) show Gaussian distribution of weeds on a simulated pasture with 40, 160 and 640 weeds respectively.

TABLE I
RANGE OF VALUES OF VARIOUS SIMULATION PARAMETERS

Parameter	Values
minimum turn radius: $\mathcal{R}(m)$	$\{1, 2, 3, 4, 5\}$
depth of detection FOV: $\mathcal{S}_d(m)$	$\{4, 8, 12, 16, 20\}$
width of detection FOV: $\mathcal{S}_w(m)$	$\{4, 8, 12, 16, 20\}$
# of weeds: $\mathcal{N}$	$\{20, 40, 80, 160, 320, 640\}$

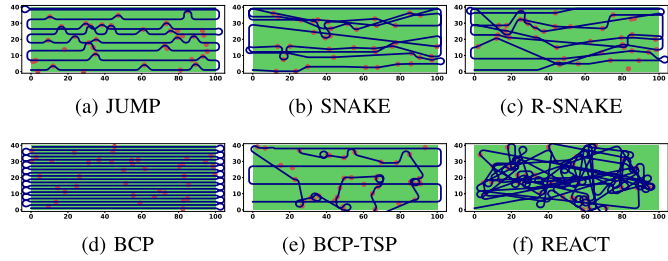


Fig. 5. Sample paths for the Cowbot planned using various planning algorithms to mow 40 weeds on a simulated pasture. Green area marks the pasture boundary, red disks mark the weeds and the solid blue line shows the mower path. (a) JUMP (b) SNAKE (c) R-SNAKE (d) BCP (e) BCP-TSP (f) REACT.

weeds $\mathcal{N} = ||\mathcal{W}||$ on the pasture (in turn the weed density), detection FOV and turning radius of the mower to study their effect on paths computed by the online planners. For each combination of weed distribution (U/G) and weed count $\mathcal{N}$, we generate 100 random instances. Table I lists the names and range of values for each of the simulation parameters.

B. Results

We use two evaluation metrics to compare and evaluate the performance of our solution methods - a) length of the mower's path and b) percentage of weeds mowed on the pasture. For ease of exposition we express the path length as a percentage of BCP path length that provides an upper bound on the solution space for MRP. We remark that (a) BCP ensures 100% weed coverage and by definition has path length equal to 100%, (b) BCP-TSP ensures 100% weed coverage on the pasture while trading-off in terms of path length and (c) REACT trades-off weed coverage in favor of fixed path length equal to 100% of the BCP length.

1) *Effect of weed density*: Density of weeds on the pasture visibly affects the path length (Fig. 6); it increases consistently with increase in the number of weeds for all planning algorithms. For smaller weed populations, each of our online planning

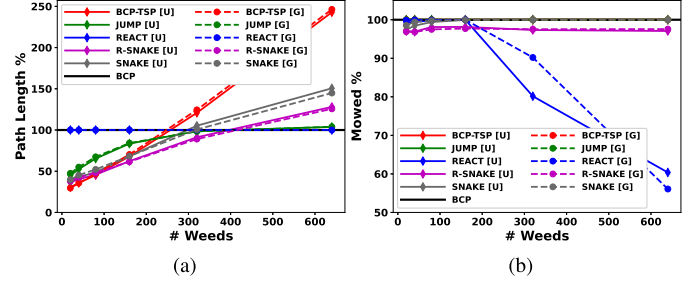


Fig. 6. The plots show trends in (a) path length and (b) % of weeds mowed, as a function of number of weeds on the pasture for various planning algorithms. Both plots show trends for both uniform and Gaussian distributions. Values for other parameters: $\mathcal{R} = 2m$, $\mathcal{S}_d = 12m$, $\mathcal{S}_w = 12m$.

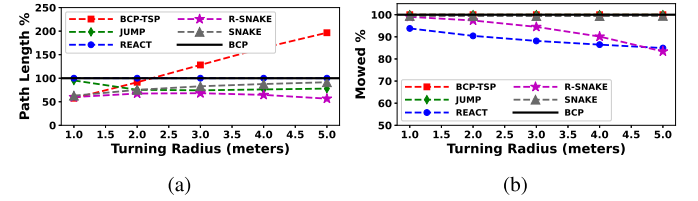


Fig. 7. The plots show trends in (a) path length and (b) % of weeds mowed, as a function of turning radius of the mower. The plots show data points averaged over all sizes of weed population. Fixed parameters: $\mathcal{S}_d = 12m$, $\mathcal{S}_w = 12m$.

algorithms - JUMP, SNAKE and R-SNAKE - leads to large improvement in path length, by up to 60%, as compared to BCP and REACT. SNAKE and R-SNAKE are best suited for low density weed populations. In scenarios, where 100% weed coverage is not required, R-SNAKE provides an efficient method to reduce path length. As the weed density increases beyond a threshold, it is more efficient to mow the entire pasture instead of visiting individual weed locations. We observe that path length for JUMP converges to BCP length and does not increase proportionally. This is expected since JUMP *shrinks* the spacing between consecutive passes to accommodate the higher density and starts to mimic a BCP path. SNAKE and R-SNAKE do not support variable spacing and hence, do not have a mechanism to accommodate the high density of weeds. Path length for these two planners increases consistently with increase in the size of the weed population.

Fig. 6(b) shows additional insights on the effects of weed density. The percentage of weeds mowed by the random-search based reactive planner (REACT) drops drastically as the weed density increases since the effect of repetitive area coverage due to random sampling becomes more pronounced. JUMP and SNAKE planners on the other hand, ensure 100% and 99.4% weed coverage, respectively. R-SNAKE planner leads to 97.5 % coverage on average, even though the detection rate for R-SNAKE is $\sim 100\%$. This is consistent with its design philosophy to trade-off weed coverage in favor of shorter path length and also explains the observation that R-SNAKE consistently has shorter path length than SNAKE.

2) *Effect of turning radius*: We make two major observations from Fig. 7(a) and (b) showing the effects of turning radius on the performance of the planning algorithms. First, the weed

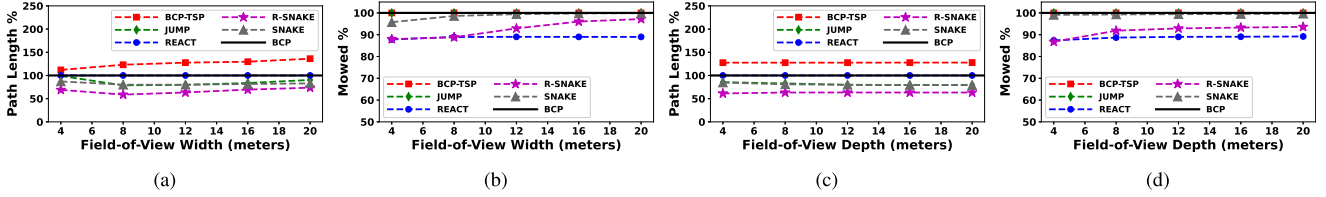


Fig. 8. The plots show trends in path length, (a) and (c), and % of weeds mowed, (b) and (d), as a function of change in FOV of the mower. (a) and (b) show variation with FOV width and (c) and (d) show variation with FOV depth. The figure plots data averaged over all $\mathcal{R}$ values. S_d was set to 12 m for (a) and (b), and S_w was set to 12 m for (c) and (d).

coverage of R-SNAKE decreases with increase in turn radius. As the mower moves to the top, weeds towards the bottom of the pasture are not considered anymore for mowing thus decreasing weed coverage. Large turn radius reduces the mower's agility and results in fewer subpaths on each pass, thus magnifying the effect of restricted range of R-SNAKE. Second, when turn radius is very small, path length for JUMP shows an abrupt increase. On investigation, we found that small turn radius makes it easier for the mower to compute jumps to mow nearby weeds. This increases the number of jumps on the mower's path, thereby increasing the path length.

3) *Effect of detection range:* Contrary to intuition, we found that depth of the FOV does not have any noticeable effect on path length (Fig. 8(c)). Width of the FOV, on the other hand, has a more pronounced effect on path length. We observe a parabolic trend in the path length for our online planners when the width of the detection FOV, S_w , is varied (Fig. 8(a)).

We note that S_w affects the detection of weeds that do not lie on the current pass. In the case of JUMP, when the FOV is narrow, the path has negligible number of jumps and converges to BCP. When the FOV is wide, the mower's path comprises of excessive number of jumps resulting in an increase in the path length. In the case of SNAKE and R-SNAKE, a narrow FOV leads to missed detection of weeds on the pasture resulting in reduced weed coverage (Fig. 8(b)). In case of R-SNAKE (Fig. 8(b) and (d)), the effect of smaller FOV due to either of S_d or S_w is magnified due to its restriction of movement, leading to missed weed detections and reduced coverage.

VIII. FIELD EXPERIMENTS

Cowbot has been tested extensively on multiple farm fields with different types of terrain. It is able to mow large and dense weed populations and is able to operate in the rugged environment on cow pastures. The custom designed navigation controller for Cowbot was tuned extensively in environments with varying topography including grass fields, steep hills, cow pastures with dense vegetation and rough terrain. The cut-quality of Cowbot and performance of the navigation controller were tested using BCP paths on cow pastures. Fig. 9 shows aerial views of a cow pasture located in Morris, Minnesota before and after a Cowbot BCP mowing experiment. To evaluate the performance of the online planning algorithms for Cowbot, field experiments were conducted on a farm field located near Morgan

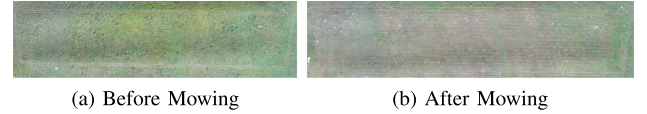


Fig. 9. Before (a) and after (b) aerial views of a cow pasture of size 90×20 sq. meters mowed by the Cowbot with a boustrophedon coverage path. The pasture boundary in the before image is highlighted by a mowed perimeter.

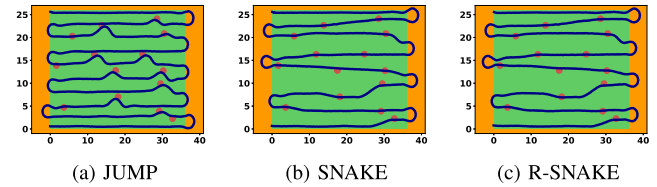


Fig. 10. Visualization of the path followed by Cowbot in field experiments using different online planning algorithms: (a) JUMP (b) SNAKE (c) R-SNAKE. Green area marks the boundary of the pasture area and red disks show the location of weed markers on the field.

in Minnesota, United States during Cowbot's public demonstrations at the Minnesota Farm Fest 2021. The experiments were conducted in an area of size 36×26 sq. meters.

For the purpose of these experiments we assume the availability of a weed detection module with a detection FOV of size $(S_d, S_w) \equiv (8, 6)$, similar in configuration to the mower model discussed in Section VI. To visualize the experiments we geotagged 14 weed locations, sampled uniformly at random, on the field and marked them with flags. Cowbot's instantaneous FOV was computed based on its location and heading estimates. When a geotagged weed entered the FOV of the mower it was marked as detected and its location information was made available to the planner. We conducted field experiments for JUMP, SNAKE and R-SNAKE planning algorithms. The BCP path length computed offline for the pasture is 897.00 m. Length of JUMP, SNAKE and R-SNAKE paths followed by the Cowbot on the field were 410.78 m (45.79%), 337.94 m (37.67%) and 338.04 m (37.68%) respectively. The numbers in bracket show the relative path length to the BCP path. Each of the mower paths was able to mow all 14 weeds on the field. Fig. 10 traces the paths followed by the Cowbot for each of the three online planners. The experiments show that Cowbot is well suited to mow weeds on pastures and our online planning algorithms are computationally efficient to deploy on real-time systems.

Experiment video showing the Cowbot in action is included in the multimedia attachment.

IX. CONCLUSION AND FUTURE DIRECTIONS

Lawn mowers and smaller robotic platforms are not well suited for operating in dairy farms. We presented the design of Cowbot, our autonomous weed mowing robot to operate in rugged terrain on cow pastures. The density and distribution of weeds on a pasture can vary significantly. As a result, coverage paths do not always provide the most efficient solution. We formulated the mower routing problem and developed online path planning algorithms, named JUMP and SNAKE (and R-SNAKE as a variant of SNAKE), to efficiently detect and mow weeds on the pasture. We deployed our online planning algorithms on the Cowbot and validated their suitability for real-time path planning in field experiments. We also performed large scale simulated experiments to compare the performance of our planners with multiple baseline methods. We conclude that SNAKE and R-SNAKE are best suited for low density weed populations and R-SNAKE provides an efficient alternative when 100% weed removal is not essential. JUMP consistently provides 100% weed coverage in all scenarios including very high weed density and shows an asymptotic increase in path length with increase in the number of weeds.

We are currently working on incorporating on board computer vision algorithms on the Cowbot to detect weeds on the fly. In our future work we would also like to address energy efficiency, robustness and safety of operating Cowbot.

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